

MYSTIC ARTS AR

Doctor Strange Augmented Reality Spellcasting Engine — User & Operation Guide

Crafted & Developed by: elsemary • © 2026 All Rights Reserved

1. Project Overview

Mystic Arts AR is a high-performance, real-time Augmented Reality web application that translates hand gestures into cinematic Doctor Strange visual spells. Built with **MediaPipe Hands**, HTML5 Canvas 2D, and vanilla JavaScript, it runs entirely client-side at a smooth 60 FPS without external heavy 3D dependencies.

2. Quick Start & Execution

Follow these simple steps to run the application locally on any computer:

Step	Action	Details
1	Launch Server	Double-click run.bat or execute python serve.py in terminal.
2	Open Browser	Navigate to http://localhost:8080 (opened automatically).
3	Camera Access	Click Allow when the browser asks for webcam permission.
4	Cast Spells	Hold your hands in front of the camera to summon mystic shields and spells.

3. Gesture & Spell Reference Guide

The system features specialized gesture classification with strict temporal validation to prevent false triggers:

Spell Name	Hand Gesture	Visual Effect & Mechanics
Eye of Agamotto (Phase 1: Forming)	Bring index & thumb tips of both hands together at chest	Summons the golden Eye relic at chest center, blazing Time Stone, rotating chrono-dials, and runic tendrils tethered to wrists.
Eye of Agamotto (Phase 2: Expansion)	Pull both hands apart horizontally from chest position	Triggers a massive expanding emerald shockwave with radial time rays, sacred tick marks, and chrono-particles.
Eye of Agamotto (Single Hand)	Pinch thumb and index fingertips together	Opens an individual emerald Eye of Agamotto with an almond frame and spinning sacred triangles.
Tao Mandala	Open palm facing camera (all fingers spread)	Authentic interlocking sacred triangles, rotating runic rings, and directional spark showers matching active theme.
Mirror Dimension	Open palm + 'Mirror' (Cyan) theme selected	Crystalline reality shatter with dynamic kaleidoscopic glass shards reflecting the environment.
Eldritch Whip	Point index finger forward (pinky & ring curled)	Electric high-voltage plasma energy whip extending directly from the fingertip with lightning branches.
Energy Fist	Clench all fingers tightly into a solid fist	Charges a concentrated spherical plasma aura over the knuckles and an eldritch wrist bracer.
Sling Ring Portal	Bring both open palms close together	Giant rotating interdimensional portal vortex generating dense bursts of swirling fiery sparks.

4. Interactive Toolbar Controls

- **Color Themes:** Switch between **Eldritch (Orange)**, **Time Stone (Green)**, **Dark Dimension (Purple)**, and **Mirror (Cyan)**.
- **Scale (+ / -):** Dynamically scale the size of the mandalas and shields from 60% to 180%.
- **Spellbook:** Opens the interactive in-app visual gesture catalog.
- **Switch Camera:** Instantly cycle between available webcams (laptop camera prioritized by default).

5. Technical Architecture

- **Offscreen Inference:** Webcam frames are rendered to an offscreen buffer (320x180) for lightweight MediaPipe detection.
- **Smooth Lerp Interpolation:** Landmark coordinates use linear interpolation (0.42 factor) to eliminate jitter.
- **Palmar Anatomical Mapping:** Real anatomical palm lines (Life Line, Head Line, Heart Line) dynamically rendered on hand bones.